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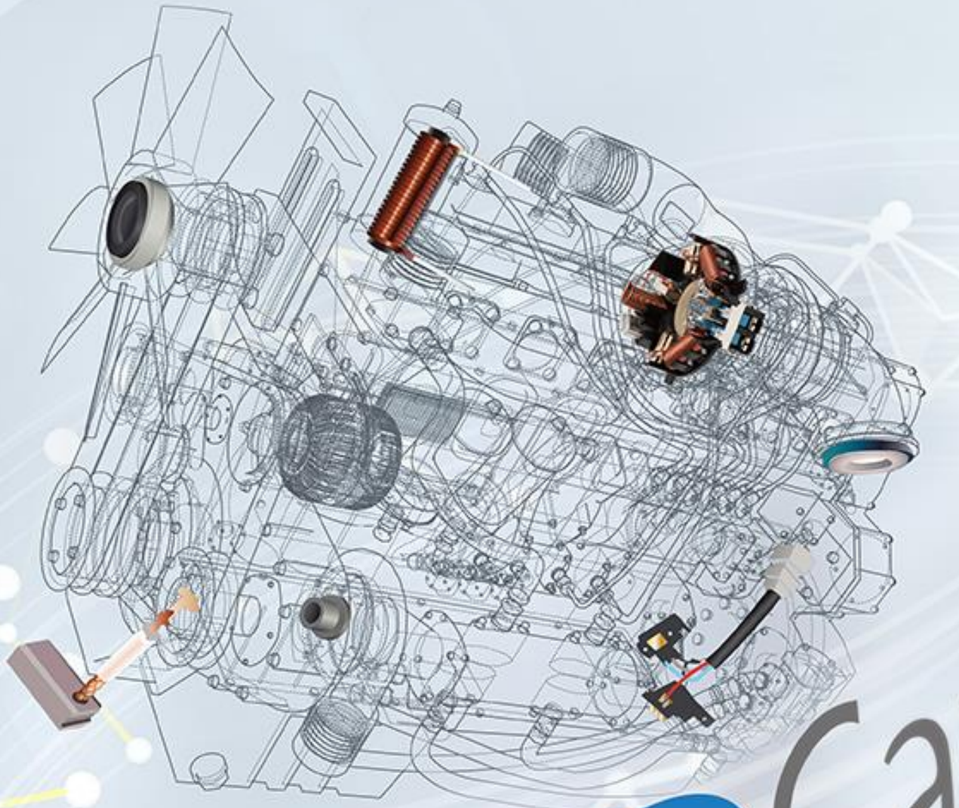

AVO Carbon GROUP

AVO Carbon Kunshan

Part No.: SAM33-DK Choke

六、点胶的措施验证 Action Verification：

固定线圈的内径，分别使用上极限和下极限的线圈对三点点胶的状况进行验证，具体如下：

Inner diameter of the coil was fixed, and the glue dispensing at three points was verified using coils at both the upper and lower limit. results are as follows:

极限磁芯

单位 mm

Core Lower Limit	3.35-3.37	Micro-meter				Micro-meter		
3.370		3.369	3.362	Core Upper Limi	3.53-3.55		Coil inner diameter	Caliper
3.362		3.371	3.362	3.529	3.530			3.59
3.373		3.368	3.364	3.534	3.529			3.57
3.369		3.370	3.366	3.530	3.528	3.524/		3.59
3.363		3.372	3.371	3.526	3.528			3.60
3.372		3.364	3.366	3.525	3.530			3.60
3.372		3.370	3.364	3.531	3.536			3.59
3.370		3.362	3.368	3.528	3.532			3.59
3.367		3.367	3.366	3.533	3.528			3.59
3.372		3.363	3.372	3.525	3.525	3.523/		3.60
3.366		3.365	3.369	3.529	3.530			3.60
3.372		3.363	3.362	3.533	3.527			3.60
3.372		3.371	3.362	3.530	3.530			3.60
3.370		3.364	3.364	3.524	3.532			3.59
3.370		3.369	3.372	3.527	3.530			3.59
					3.533			3.59
Max	3.373	3.372	3.372	3.534	3.530			
Min	3.362	3.362	3.362	3.524	3.530			
					3.530			
					3.536			
Gap					3.525			
	Min Gap	0.197	0.197					
	Max Gap	0.238	0.238	Gap				
							Min Gap	0.034
							Max Gap	0.076

验证结果Verification Results

Gluing method: Three points Attendant: Vivian Ms Feng Ms Cui

Gap(mm)	Gluing method	Glue Quantity	Force test						Biting status	Note
0.197-0.238	Three point	2.8mg φ3.35				197.4	126.4	227.4		
	Three point	3.0mg φ3.35				469.5	392.8	401.4		
	Three point	5.1mg φ3.35				557.7	447.7	377.2		
	Three point	6.5mg φ3.35				666	508.8	681.7		有轻微溢胶但不超出0.3mm
	Three point	7.0mg φ3.35		> 1000N	> 1000N	1262.7	> 1000N	> 1000N		有轻微溢胶但不超出0.3mm
0.034-0.076	Three point	3.0mg φ3.55				838.7	401.4	469.5		
	Three point	5.1mg φ3.55	> 1000N			> 1000	1007.6	898.6		有轻微溢胶但不超出0.3mm
	Three point	6.5mg φ3.55		> 1000N	> 1000N	1131.1	> 1000N	> 1000N		有明显溢胶现象，咬合力依靠的是堆积在磁芯端部的胶水；大于6.5mg胶水没有再进行验证
Result	根据以上验证：根据产品外观状况，产品推脱力和设备状况等，胶量锁定为3-7mg;									

根据以上验证结果，三点点胶的推脱力完全满足要求，且根据产品外观，推脱力以及出胶状况胶量控制在3-7mg；

1. 磁芯在下限时，胶量越多，推脱力越大；胶量和推脱力正相关，胶量在上限时易出现溢胶现象，但不超过0.3mm。
2. 磁芯在上限时，胶量越大，推脱力越大；推脱力和胶量正相关；磁芯在上限且胶量为5.1 mg出现溢胶现象，但不超过0.3mm；6.5mg时出现明显溢胶。
3. 胶量固定的状态下，间隙和推脱力成负相关。

Verification results: Push test & appearance meet the requirements when the glue weight is between 3-7mg;

1.When core diameter is at lower limit, the more glue, the greater push-off force; the glue volume VS push-off force are positively correlated. When the glue volume is at upper limit, glue overflow to occur, but it does not exceed 0.3mm.

2.When the core is at upper limit, the more glue volume, the greater push-off force; push-off force VS glue volume are positively correlated. When the core is at upper limit, the glue volume is 5.1mg, glue overflow occurs, but it does not exceed 0.3mm; when the glue volume is 6.5mg, glue overflow occurs obviously.

3.At a fixed glue volume, the gap and push-off force are negatively correlated.

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七、系统的措施Systematic measures taken：

- 更新控制计划中推脱力的检查频次和要求 （before：1次/shift After: 1次/2小时）

  责任人：Vivian Wang　　要求完成时间：9月26日

- 更新PFMEA

  责任人：Ding　　要求完成时间：9月28日

- 推力测试机力值点检

  责任人：Sunny　　要求完成时间：9月26日

- 对相关人员进行培训

  责任人：Zhuzhu　　要求完成时间：9月26日

- 横向展开检查其它手工点胶产品是否存在类似状况

  责任人：Ding & Leaf　　要求完成时间：10月12日

- 经验教训总结: 昆山手动点胶均已经改善点胶方式

  责任人：Vivian Wang　　要求完成时间：10月12日

•Update inspection frequency for push test in control plan (Before: 1 time/shift, After: 1 time/2 hours)
Resp.: Vivian Wang Date: September 26th

•Update the PFMEA
Resp.: Ding　Date: Sep. 28th

•Thrust test machine force value inspection

•Resp.: Sunny　Date: Sep. 26th

•Train relevant personnel
Resp.: Zhuzhu Date: Sep. 26th

•Horizontally expand the inspection to check if similar conditions exist in other manual products
Resp.: Ding & Leaf　Date: Oct. 12th

•Lessons learned;

Resp.: Vivian Wang Date: Oct. 12th

备注：跟踪客户端反馈，从9月26日至10月15日，共计出货超出1200K PCS，类似问题没有再次发生，措施有效，问题结案。
Note: Have tracked the client's feedback, From October 15th to December 30th, a total of over 400,000 pieces were shipped and used, the problem was not re-occurred. The actions were effective and it is closed.

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